

Part 3: Damping Factor β

Test Case:

Mass m (kg)	0.05
Initial amplitude A_1	0.016
Amplitude after t: A_2	0.004
Time interval t (s)	63
$\ln(A_1/A_2)$	1.386294361
$\beta = \ln(A_1/A_2)/t$ (s^{-1})	0.022004672

Trial	t (s)	β (s^{-1})	error (s^{-1})
1	100	0.02390	0.0002877
2	100	0.01928	0.0006276
3	100	0.02231	0.0004106
4	100	0.01863	0.0009143
5	100	0.02356	0.0003986

Mean β (s^{-1})	0.02154
Std error β	0.00109
$\beta \pm SE$ (s^{-1})	0.022 $\pm$ 0.001